PawBridge – A Digital Bridge for Animal Welfare

-“Bridging compassion and care for every paw”

Problem Statement

Many of stray animals suffer injuries, neglect, and abuse.

Citizens lack a **single, accessible platform** to report and help.

Existing welfare efforts are scattered, causing **delays and inefficiency.**

Objective

Enable **real-time reporting** of injured or homeless animals.

Connect citizens to **nearby clinics, NGOs, and shelters**

Promote **adoptions, volunteering** and **report abuse** with geo-tagged evidence.

Facilitate **donations** with transparency.

Expected Contributors

Social Impact: Faster rescue, reduced suffering, citizen empowerment.

Technological Impact: Demonstrates real-time, geo-enabled rescue platform.

Scalability: Extendable to mobile apps, government integration, national NGO networks

Tech Stack

Frontend: HTML, CSS and JavaScript

Backend: Node.js and Express.js

APIs: Leaflet and OpenStreetMap

Hosting & Tools: GitHub, Postman and Visual Studio Code

Methodlogy

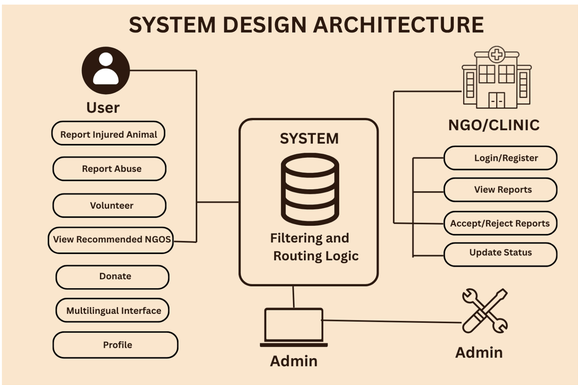
Report → User uploads photo, injury details, GPS location.

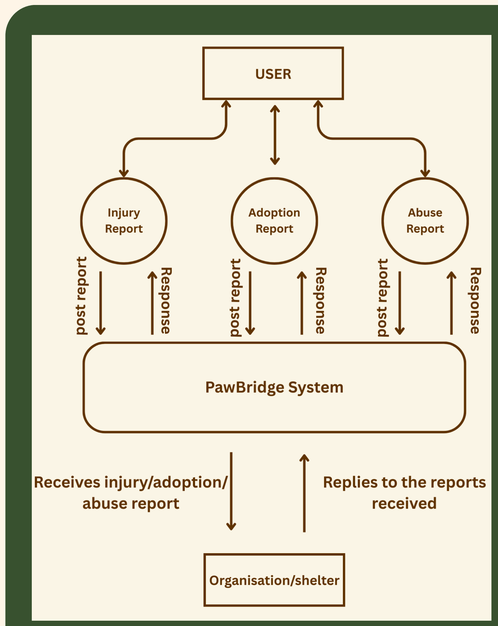
Locate → System suggests nearby clinics/NGOs via Maps API.

Connect → Notifications sent to relevant rescuers/shelters.

Support → Donation portal , volunteering opportunities and adoption listings.

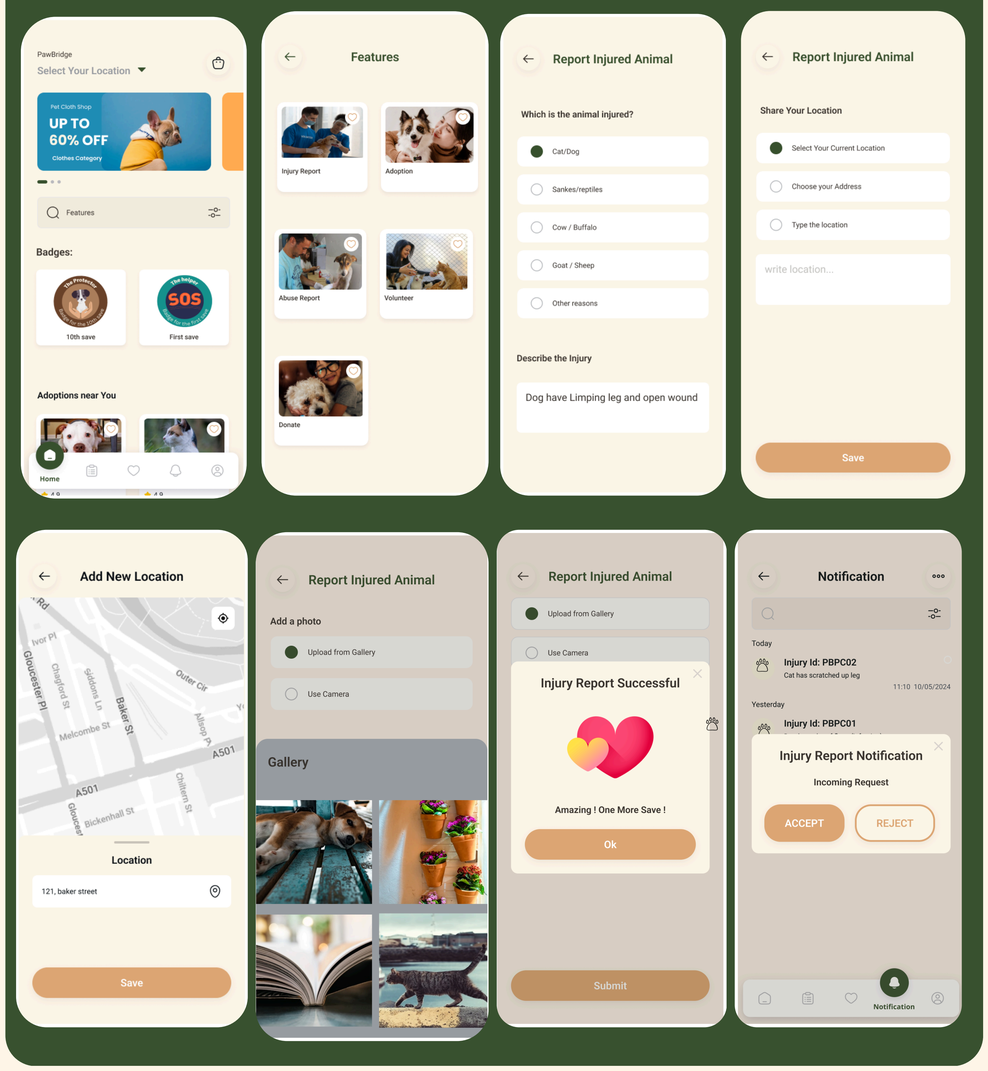
Monitor → Role-based dashboards for tracking & management.





UI/UX :

The images show the user interface of the PawBridge app. These screens show the report of an injured animal being sent to the NGO/Organisation/Animal Shelter



Future Scope

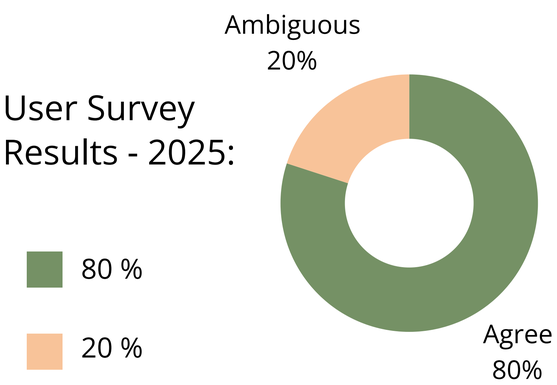
CNVR Programs inculsion

AI for image-based injury detection & report duplication

Onboarding municipal corporations

Analytics

User Survey Results - 2025:



Conclusion

PawBridge merges compassion with technology to create a reliable, scalable, and community-driven solution for animal welfare. It transforms ordinary citizens into active rescuers, ensuring timely help and a kinder future for animals

References

Humane Society International:Technology for improving efficiency and welfare during street dog sterilization

WILET(Online Library):- Advancements in artificial intelligence technology for improving animal welfare: Current applications and research progress

SSRN:-Smart Surveliance and animal care system using IOT and deep learning